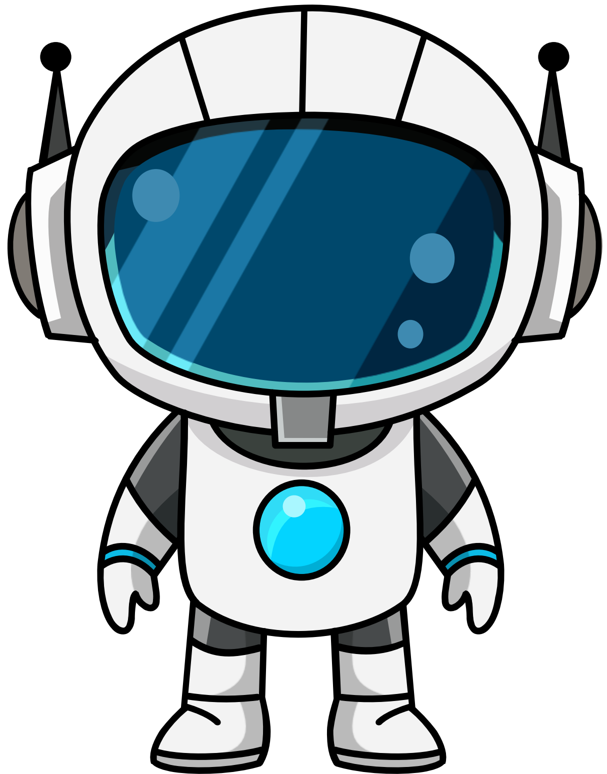
KATALYST TECH FEST 2026

PsycheScan-AI Emotion Recognisation System



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PROBLEM STATEMENT

Importance of Problem

•Emotional distress often goes unnoticed in digital environments.
•Lack of real-time tools to analyze emotions during interactions.
•Need for intelligent systems to understand human emotions.

What We Provide

•AI-based system to detect human emotions.
•Uses facial expressions
•Real-time emotion recognition.
•Dashboard for insights and analysis.

OUR SOLUTION

• Real-time emotion detection using AI and computer vision
• Facial expression analysis using deep learning (CNN-based model)
• Live camera input processing for instant emotion recognition
• Emotion classification (Happy, Sad, Angry, Neutral, Surprise, etc.)
• Interactive dashboard showing emotion insights and results
• Fast and efficient processing for smooth user experience

IMPACT & BENEFITS

IMPACT

- Enables early detection of emotional states for better mental health awareness
- Improves human-computer interaction through emotion-aware AI systems
- Enhances engagement analysis in education, healthcare, and workplace environments
- Supports data-driven decision making using emotional insights

BENEFITS

- Real-time and accurate emotion recognition
- User-friendly and efficient monitoring system
- Privacy-aware and scalable AI solution

TECHNOLOGIES USED

Tech-Stack

Web App-**HTML, CSS, JAVASCRIPT**

Backend / API- **Flask & Python**

Machine Learning (SSE)- **TensorFlow / Scikit-learn**

Database-**MongoDB**

Dataset: **FER2013**

AI/ML Frameworks: **TensorFlow / Keras / PyTorch**

End-to-End System Workflow Architecture

PRESENTATION LAYER

- User Interface (HTML, CSS, JavaScript)
- Login / Signup
- Dashboard
- Live Video Detection
- Upload Recorded Video

APPLICATION LAYER

- Flask Backend Server
- Route Handling
- Session Management
- Video Processing Control
- Model Invocation

AI PROCESSING LAYER

Face Detection Module (OpenCV + Haar Cascade)

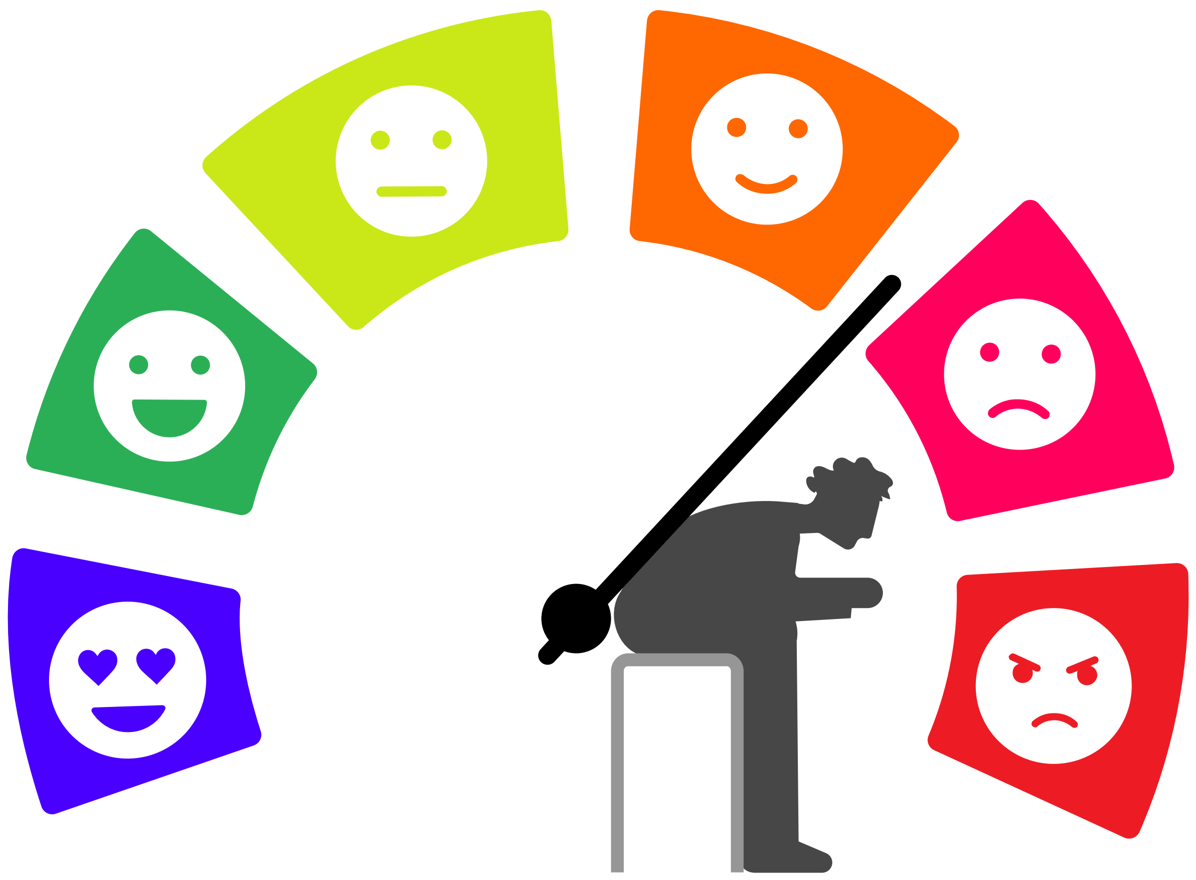
Emotion Classification Model (CNN - TensorFlow / Keras)

Emotion Analytics Engine

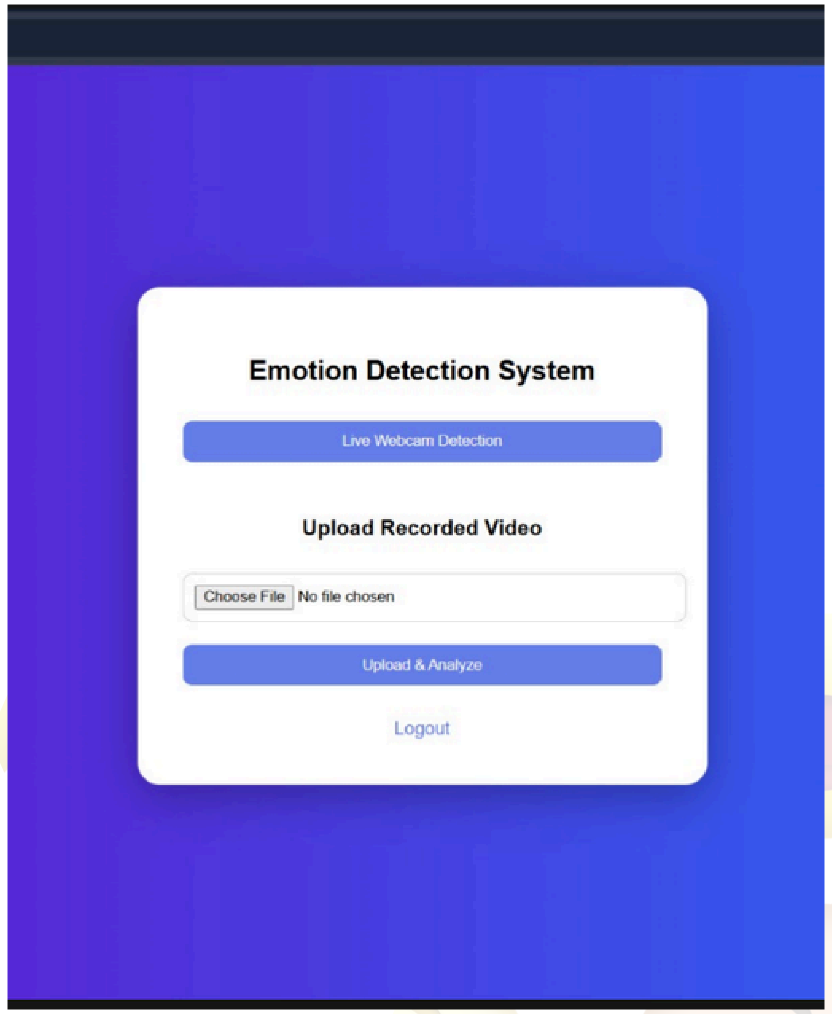
- Emotion Percentage Calculation
- Valence Classification
- Stability Score Computation
- Final Decision

OUTPUT LAYER

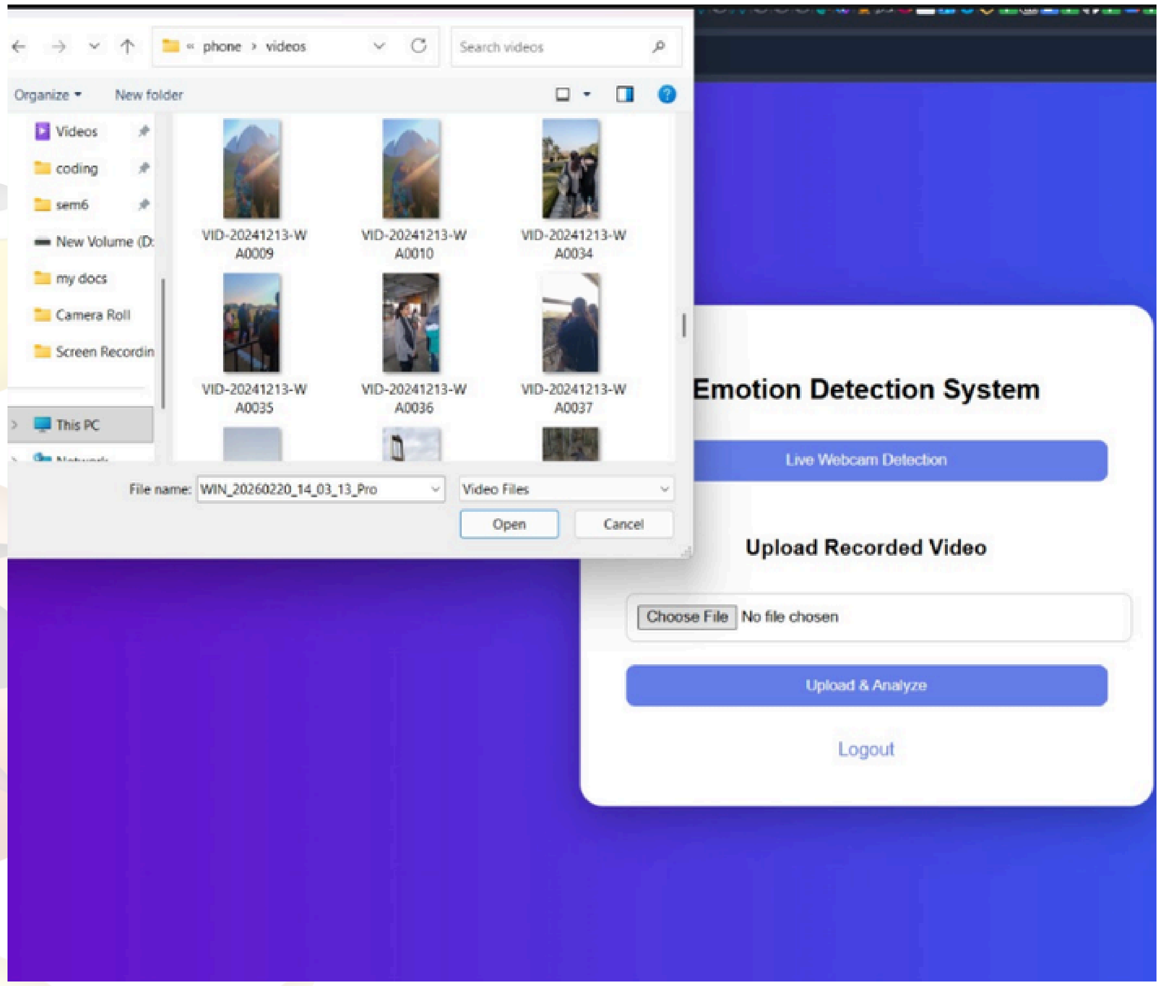
- Report Generation Module
- Txt Report
- Download Report Option



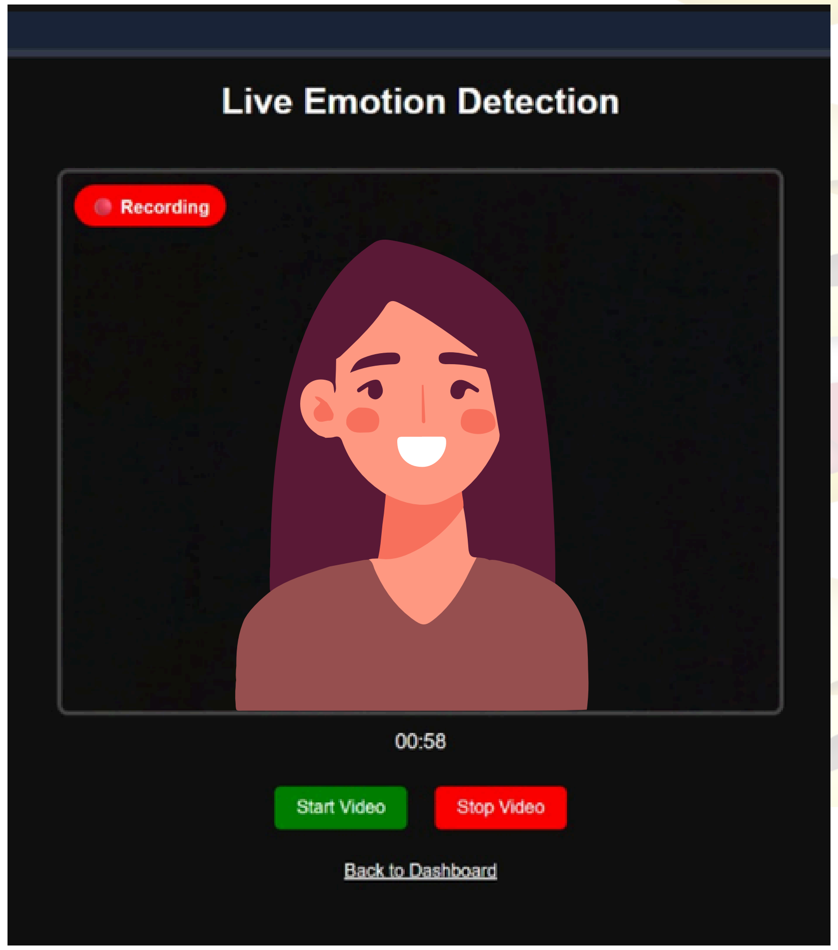
Emotion Detection System UI



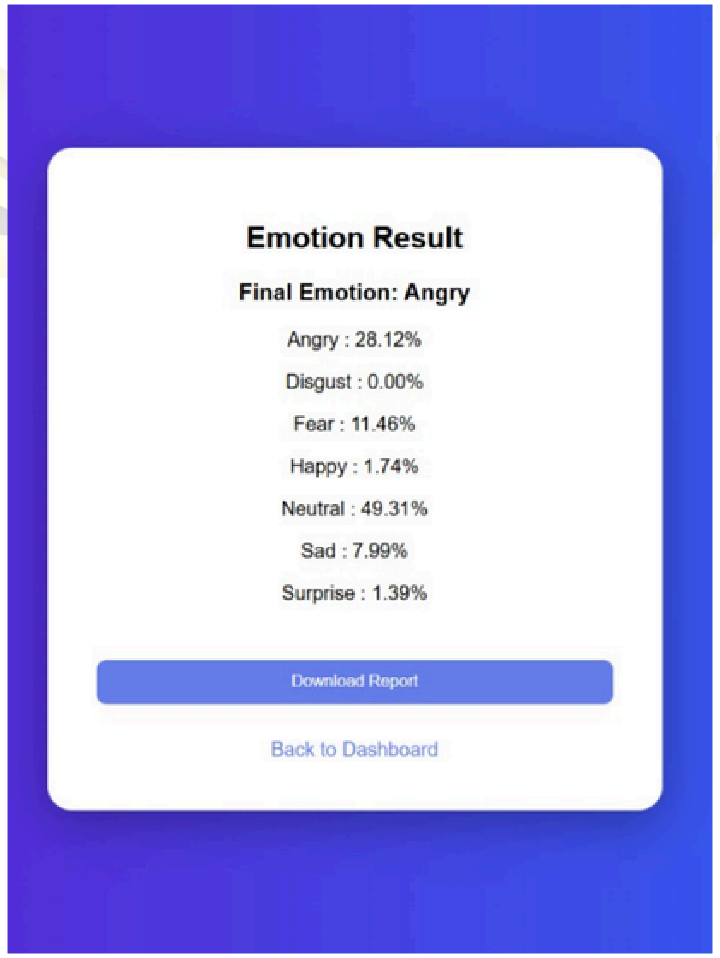
Dashboard / Home Screen



Upload Recorded Video



Live Emotion Detection



Emotion Analysis Result

CNN MODEL ARCHITECTURE

Data Preprocessing Stage

Dataset

- FER2013 dataset
- 7 emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral

Image Processing

- Resize images to 48×48 pixels
- Convert to grayscale
- Normalize pixel values (0–1)

Cleaning & Validation

- Removed duplicate images across classes
- Checked for corrupt/unreadable images

Data Augmentation

- Random horizontal flips
- Random rotation (±10%)
- Random zoom (±10%)

CNN MODEL ARCHITECTURE

CNN Model Training

Model Architecture

- Input: 48×48×1 grayscale images
- 3 Convolutional Blocks
- Conv2D ×2 → BatchNorm → MaxPool → Dropout
- Filters: 64 → 128 → 256
- Fully Connected Layer: 512 neurons → BatchNorm → Dropout 0.5
- Output: Dense 7 → Softmax

Training Details

- Optimizer: Adam
- Loss: Categorical Crossentropy
- Metrics: Accuracy
- Epochs: 100
- Batch size: 128
- Callbacks: EarlyStopping, ModelCheckpoint

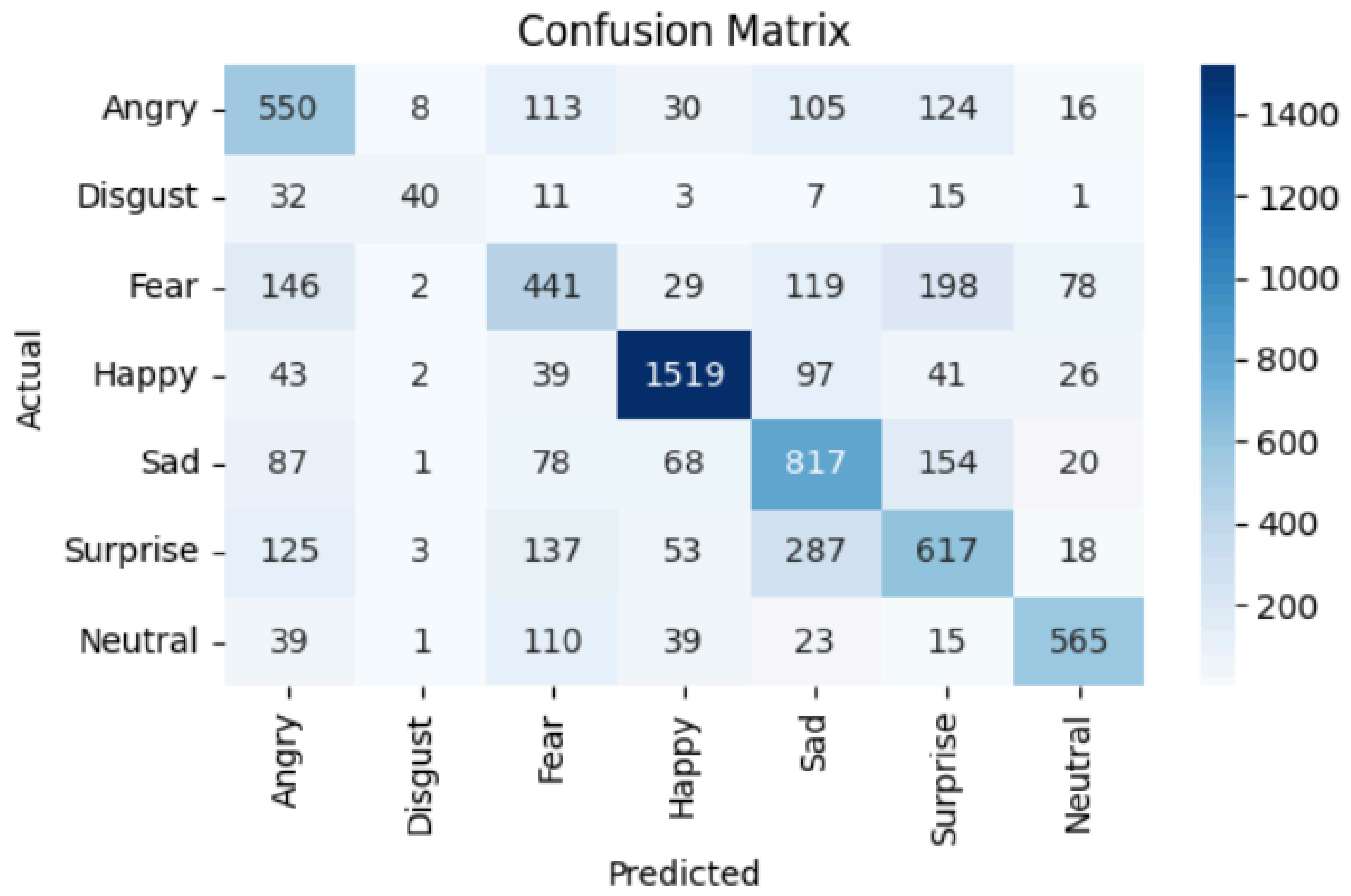
Training & Validation Performance

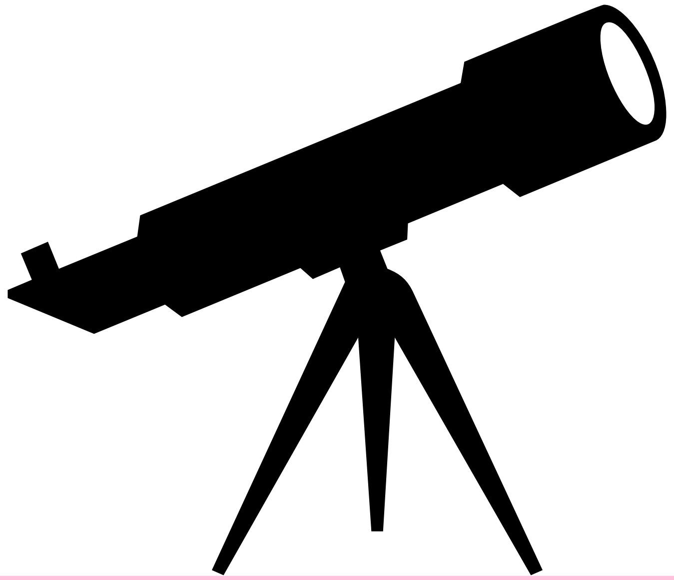
- Model training reached train accuracy = 70%
- Validation accuracy = 59%
- EarlyStopping used to prevent overfitting after 47 epochs
- Confusion Matrix to analyse true prediction and false prediction.

```
Epoch 4: finished saving model to cnn_first_model.keras
211/211 ━━━━━━━━━━━━━━━━━━━━ 426s 2s/step - accuracy: 0.5376 - loss: 1.2074 - val_accuracy: 0.5677 - val_loss: 1.1311
Epoch 5/100
...
Epoch 47/100
211/211 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.7073 - loss: 0.7855
Epoch 47: val_accuracy did not improve from 0.64862
211/211 ━━━━━━━━━━━━━━━━━━━━ 666s 3s/step - accuracy: 0.7040 - loss: 0.7908 - val_accuracy: 0.5917 - val_loss: 1.1575
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Layer (type)	Output Shape	Param #
sequential (Sequential)	(None, 48, 48, 1)	0
conv2d (Conv2D)	(None, 48, 48, 64)	640
batch_normalization (BatchNormalization)	(None, 48, 48, 64)	256
conv2d_1 (Conv2D)	(None, 48, 48, 64)	36,928
batch_normalization_1 (BatchNormalization)	(None, 48, 48, 64)	256
max_pooling2d (MaxPooling2D)	(None, 24, 24, 64)	0
dropout (Dropout)	(None, 24, 24, 64)	0
conv2d_2 (Conv2D)	(None, 24, 24, 128)	73,856
batch_normalization_2 (BatchNormalization)	(None, 24, 24, 128)	512
conv2d_3 (Conv2D)	(None, 24, 24, 128)	147,584
batch_normalization_3 (BatchNormalization)	(None, 24, 24, 128)	512
max_pooling2d_1 (MaxPooling2D)	(None, 12, 12, 128)	0
dropout_1 (Dropout)	(None, 12, 12, 128)	0





FUTURE SCOPE

Development of a mobile application for better accessibility and user engagement

Implementation of real-time analytics dashboard for data visualization

Cloud deployment for scalability, reliability, and secure data storage

Multi-user role management with advanced authentication & authorization

Performance optimization and security enhancements (encryption, data validation)

Expansion of features based on user feedback and real-world testing

THANK YOU